

ERIC TAL MYZELEV

EDUCATION

Carnegie Mellon University

August 2025 - Present

- PhD in Algorithms, Combinatorics, and Optimization
- Advisor: Dr. Maria-Florina Balcan
- GPA: 3.96/4.00
- Relevant Coursework: Machine Learning Theory; Introduction to Deep Learning; Algorithms in the Real World; Linear Programming; Probability; Discrete Math; Extremal Combinatorics

University of Pennsylvania

August 2021 - May 2025

- Degrees: Master of Arts, Bachelor of Arts
- Major: *Mathematics*, Minor: *Computer and Information Science*
- Master's GPA: 4.00/4.00, Bachelor's GPA: 3.96/4.00

RESEARCH EXPERIENCE

Machine Learning Theory Research

November 2025 – Present

Mentor: Professor Nina Balcan, Carnegie Mellon University

- Studying the complexity of hyperparameter tuning in neural networks
- Proved VC-dimension lower bounds for the hypothesis class induced by tuning the Leaky ReLU slope parameter
- Showed that even when only this single hyperparameter varies, the induced class has VC dimension at least linear in the network width

REU at Carnegie Mellon University: Geometry and Topology in a Discrete Setting

May 2024 – July 2024

Mentor: Professor Florian Frick, Carnegie Mellon University

- Developed methods to bound distances between successive roots in symmetric maps using equivariant topology
- Obtained a topological proof of a theorem of Babenko about roots of trigonometric polynomials

Auburn University REU in Discrete Math and Algebra

May 2023 – July 2023

Mentor: Professor Peter Johnson, Auburn University

- Proved an equivalence between the Hadwiger-Nelson problem and a hypergraph generalization
- Characterized colorings of $\mathbb{R}^d$ produced by Szlam's Lemma

Research Assistant in Physics-Informed Neural Networks Lab

May 2022 – May 2023

Mentor: Professor Lu Lu, University of Pennsylvania

- Developed an adaptive domain decomposition technique to improve PINN accuracy for stiff differential equations
- Worked on applications of PINNs to hemodynamics and blood clot modeling using Navier-Stokes equations and MRI data
- Presented this work at the University of Pennsylvania Undergraduate Research Expo

PUBLICATIONS

1. Superpolynomial lower bounds for vertex numbers of real projective space triangulations via a topological Figiel-Lindenstrauss-Milman theorem
with Florian Frick, Kaave Hosseini, Arya Narnapatti, Aliaksei Vasileuski
Preprint
<https://arxiv.org/abs/2609.10402>

2. Roots of Real-Valued Zero Mean Maps
with Francesca Cantor, Julia D'Amico, and Florian Frick
European Journal of Mathematics (2025)
<https://link.springer.com/article/10.1007/s40879-025-00870-5>
3. Characterization of Colorings Obtained by a Method of Szlam
Geombinatorics Quarterly, 34(4), 147–152 (2024)
<https://arxiv.org/abs/2411.04346>
4. A New Class of Geometric Hypergraphs Arising from the Hadwiger-Nelson Problem
with Sean Fiscus and Hongyi Zhang
Geombinatorics Quarterly, 34(3), 97–106 (2024)
<https://arxiv.org/abs/2411.05931>
5. Transfer Learning on Physics-Informed Neural Networks for Tracking the Hemodynamics in the Evolving False Lumen of Dissected Aorta
with Mitchell Daneker, Shengze Cai, Ying Qian, Arsh Kumbhat, He Li, and Lu Lu
Nexus, 1(2) (2024)
<https://www.sciencedirect.com/science/article/pii/S2950160124000147>

TALKS AND POSTERS

- *A New Equivalence in Euclidean Ramsey Theory*. Joint Mathematics Meetings, Pi Mu Epsilon, January 5, 2024.
- *A New Class of Geometric Hypergraphs Arising from the Hadwiger-Nelson Problem*. Southern Africa Mathematical Sciences Association Annual Conference, University of Pretoria, November 21–24, 2023.
- *Physics-Informed Neural Networks for Tracking the Hemodynamics of Dissected Aorta*. Undergraduate Research Expo, University of Pennsylvania, September 18, 2023.
- *Solving Stiff ODEs with Physics Informed Neural Networks*. Undergraduate Research Expo, University of Pennsylvania, September 19, 2022.

AWARDS

- NSF Graduate Research Fellowship Program April 2025
- Barry Goldwater Scholarship March 2024
- Phi Beta Kappa Junior Inductee May 2024
- NSF Research Experience for Undergraduates Grant May 2023, May 2024

TEACHING AND MENTORSHIP

- Mentor at Lumiere Education May 2026 - Present
- Linear Algebra Teaching Assistant, University of Pennsylvania August 2024 – May 2025
- Contest Writer for Canadian Astrophysics and Astronomy Olympiad Feb. 2021 – May 2024

SKILLS

- **Programming:** Python, Java, R, OCaml
- **Software:** TensorFlow, LaTeX
- **Languages:** English (native), Russian (fluent), Ukrainian (spoken comprehension)